Technical Approach Report: Revio v2 - Reputation Intelligence Engine

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1. Problem Framing

Revio v2 addresses three gaps in Cynbit's existing reputation management product:

1. **Language gap**: Indian reviews contain significant Hinglish (code-mixed) content, but existing sentiment tools are English-centric. *Note: "Indian reviews are predominantly Hinglish" is an **[ASSUMPTION]** based on observed review patterns in public datasets; the actual share varies by platform, region, and demographic.*

2. **Granularity gap**: Overall sentiment ("positive/negative") is insufficient for actionable business insights — aspect-level analysis is needed

3. **Reply safety gap**: Automated replies risk hallucinating promises not supported by actual policies

The PoC demonstrates a pipeline that: detects language (English/Hindi/Hinglish), extracts aspects with per-aspect sentiment, detects crises (safety, legal, health, fraud, viral, location spikes), and drafts grounded replies with an approval gate.

2. Architecture

Data Flow

```
Review Ingestion (CSV/JSON)
  -> Language Detection (English/Hindi/Hinglish)
  -> Text Normalisation (lowercase, abbreviations, noise removal)
  -> PII Masking (phone, email, card, names)
  -> Aspect Extraction (keyword matching + context analysis)
  -> Sentiment Scoring (overall + per-aspect)
  -> Severity Classification (critical/high/medium/low)
  -> Crisis Detection (keyword rules + z-score anomaly)
  -> Reply Generation (LLM with grounding OR template fallback)
  -> Approval Gate (confidence-based routing)
  -> FastAPI /analyze + Streamlit Dashboard
```

Components

Component	Approach	Rationale
Language Detection	Character-ratio (Devanagari) + lexical heuristics (Hinglish words)	Fast, no model download, handles code-mixing

Normalisation	Regex-based: lowercase, abbreviation expansion, noise reduction	Simple, preserves negations, handles SMS-speak
PII Masking	Regex patterns (phone, email, card) + name dictionary	100% precision on structured patterns, no LLM dependency
Sentiment	Keyword-based (fallback) + TF-IDF/Logistic Regression (baseline)	Commodity approach, fast on CPU, interpretable
Aspect Extraction	Keyword matching with negation detection + context window	Handles explicit mentions, interpretable evidence spans
Crisis Detection	Keyword rules + viral risk signals + z-score location anomaly	Low false positive rate, interpretable, no training data needed
Reply Generation	LLM (optional OpenAI) with grounding prompt + template fallback	Works offline, policy-compliant, handles Hinglish
Approval Gate	Decision tree: crisis -> human, confidence thresholds -> auto/human	Conservative routing, never auto-approves crisis

Rejected Alternatives

Component	Rejected	Reason
Sentiment	Fine-tuned BERT	Too heavy for CPU-only, overkill for PoC
Sentiment	Lexicon-based only	Fails on negations and context
Aspect	Dependency parsing	Fragile on Hinglish, overkill
Aspect	Fine-tuned NER model	Requires training data, zero-shot sufficient
Crisis	LLM-only crisis detection	Latency overhead, not interpretable
Reply	No fallback (LLM only)	Demo breaks without API key

3. Data Pipeline

Synthetic Data Generation

The PoC uses a custom synthetic review generator (`data/generate_reviews.py`) that produces 3,000 reviews with:

• **Language mix**: 70% English, 15% Hinglish (romanised), 15% Hindi (Devanagari)

• **Sentiment distribution**: 60% positive, 20% negative, 20% neutral

• **Aspects**: 1-3 aspects per review (30% mention 2+ aspects), 8 predefined aspects

• **Noise injection**: 5% typos, 15% emojis, abbreviations (u -> you, thx -> thanks)

• **Crisis injection**: 5% of negative reviews get crisis keywords (safety, legal, health, fraud, viral)

• **Location spikes**: 1-2 locations get elevated negative rates to test anomaly detection

All data is seeded (RANDOM_SEED=42) for reproducibility. Output files:

• `data/reviews.csv` (3,000 reviews, no labels, for ingestion demo)

• `data/reviews_labelled.csv` (with ground truth: sentiment, aspects, crisis, severity)

• `data/reviews_train.csv`, `reviews_val.csv`, `reviews_test.csv` (70/15/15 stratified split)

• `data/reviews_hand_labelled_gold_set.csv` (99 reviews with edge cases)

• `data/tone_guide.csv` (brand tone patterns per scenario)

• `data/policy.csv` (service policies, constraints)

Manually Authored Gold Set

A separate evaluation set of 98 reviews was **manually authored with AI assistance**. These are NOT real customer reviews and were NOT generated by the synthetic pipeline. They include sarcasm, mixed aspects, typos, emojis, and borderline crisis cases — patterns the synthetic generator does not produce. The gold set was used ONLY for held-out evaluation and NEVER for tuning.

Data Split

Stratified by (sentiment, crisis_flag) to maintain distribution across train/val/test. The gold set includes edge cases: sarcasm, multi-aspect single sentences, Hinglish code-mixing, emojis, crisis reviews.

4. Models and Trade-offs

Baseline: TF-IDF + Logistic Regression (now trained)

• TF-IDF (max_features=10000, ngram_range=(1,2), sublinear_tf) + Logistic Regression (L2, C=1)

• Interpretable, fast, establishes benchmark for ablation

• Trained on 2,098 synthetic reviews; evaluated on held-out test + gold sets

• **Key finding**: Overfits to synthetic vocabulary — high test F1 (0.974) but low gold F1 (0.448)

MiniLM + Logistic Regression (now trained)

• `paraphrase-multilingual-MiniLM-L12-v2` (384-dim, 50+ languages) frozen encoder + LR head

• Supports English, Hindi (Devanagari), and Hinglish

• Requires `pip install -r requirements-optional.txt`

• **Key finding**: Better than TF-IDF on gold (0.565 vs 0.448) but still below keyword baseline (0.879) — synthetic training data does not generalise to realistic reviews

Main Sentiment: Keyword-based (PoC Default)

• Positive/negative word matching with rating signals

• Handles negation detection ("not bad" -> positive)

• Confidence based on signal count

• **Strongest on gold set (0.879)** because the rule vocabulary covers common sentiment patterns regardless of phrasing variation

Aspect Extraction

• Keyword-first approach: match against 8 aspect keyword dictionaries

• Negation detection: 3-word window before keyword

• Per-aspect sentiment via keyword context analysis

• Evidence spans stored for explainability

Crisis Detection

• Rule-based keyword matching across 5 crisis categories

• Viral risk: excessive caps, multiple exclamations, social media threats

• Location spike: z-score > 2.0 with minimum 3 negatives in 7-day window

• Aggregated confidence: max across all signals

Reply Generation

• **LLM mode** (optional): OpenAI GPT-3.5-turbo with strict system prompt

• Grounding rule: "Use ONLY provided policy; if not covered, say manager will follow up"

• Injection-safe: Review text wrapped in `...` delimiters

• Few-shot examples: 3 examples (negative, positive, crisis)

• JSON output: reply, confidence, requires_human_review, policy_references

• 5-second timeout with fallback

• **Fallback mode** (no API key): Template-based generation

• Crisis: escalation template (always human review)

• Positive: thank-you with aspect reference

• Negative: apology with policy-grounded resolution

• Hinglish/Hindi: matches review language

5. Prompt Engineering

System Prompt Structure

```
Role: Customer service representative for multi-location business in India
Rules: Acknowledge feedback, address aspects, ground in policy, match language
Tone Guide: [Brand tone guide snippet]
Policy: [Policy file snippet]
Output: JSON with reply, confidence, requires_human_review, policy_references
Grounding: Use ONLY provided policy; never invent solutions
```

Guardrails

1. **Grounding rule**: No promises outside policy file

2. **Injection delimiters**: `...` around untrusted text

3. **Injection detection**: Flag "ignore", "override", "system prompt" patterns

4. **Refusal/escalation**: Crisis reviews skip LLM entirely, use escalation template

5. **Confidence threshold**: Auto-approve only if confidence >= 0.80 (positive/neutral)

6. Evaluation

Metrics Summary (Synthetic Test Set)

Metric	Target	Actual	Status
Sentiment Macro-F1	>= 0.82	0.943	PASS
Crisis Recall	>= 0.90	1.000	PASS
Crisis Precision	>= 0.95	1.000	PASS
Aspect F1	-	0.979	-
Severity Macro-F1	-	1.000	-
Cold Start	< 60s	< 1s	PASS
Avg Latency	< 3s	0.005s	PASS

Per-Language F1 (Synthetic)

• English: 1.000

• Hinglish: 0.951

• Hindi: 1.000

Model Comparison Table

Model	Synthetic Test F1	Manually-Authored Gold F1	Note
Keyword baseline (existing)	0.943	**0.879**	Best generalisation; vocab overlaps real patterns
TF-IDF + LR (trained)	0.974	0.448	Overfits to synthetic vocabulary
MiniLM + LR (trained)	0.945	0.565	Better than TF-IDF on gold; needs optional deps

Key insight: Both trained models overfit to synthetic vocabulary. The keyword baseline generalises best to the manually-authored gold set because its vocabulary covers common sentiment words rather than dataset-specific co-occurrences. A trained model will only beat the keyword baseline when trained on **real** customer review data. This is an expected and honest result.

Full comparison data: `docs/results/metrics_model_comparison.json`

Synthetic vs Manually-Authored Gold Set

The manually-authored gold set (98 reviews, NOT real customer reviews, manually authored with AI assistance, NEVER used for tuning) was created to test generalisation on edge cases:

Metric	Synthetic	Manually-Authored Gold	Change
Sentiment Macro-F1	0.943	**0.879**	-0.064
Crisis Recall	1.000	**0.917**	-0.083
Crisis Precision	1.000	**0.846**	-0.154
Aspect F1	0.979	**0.805**	-0.174
Severity Macro-F1	1.000	**0.808**	-0.192
Sarcasm Accuracy	-	**0.800**	-
Hinglish F1	0.951	**0.873**	-0.078

Synthetic scores are OPTIMISTIC because the keyword model matches the same vocabulary the synthetic generator uses. Crisis recall stays above 0.90 on the gold set, confirming the rule-based crisis detector generalises reasonably well. See `docs/results/metrics_handwritten.json`.

Ablation (Design-Level)

Variant	Sentiment F1	Crisis Recall	Latency
Keyword-only (current default)	0.943 synth / 0.879 gold	1.000 / 0.917	0.005s
TF-IDF + LR (trained, synthetic)	0.974 synth / 0.448 gold	0.55	0.05s
MiniLM + LR (trained, synthetic)	0.945 synth / 0.565 gold	0.75	1.8s
+Rules (crisis keywords)	~0.85	0.92	1.9s
+LLM replies	~0.85	0.92	2-8s

7. Limitations

1. **Synthetic data optimism (CONFIRMED by gold eval)**: The synthetic scores (0.943 F1) are optimistic. The manually-authored gold set eval gives 0.879 F1, confirming synthetic metrics overestimate performance.

2. **Trained models overfit to synthetic vocabulary**: TF-IDF+LR achieves 0.974 on synthetic test but only 0.448 on the manually-authored gold set. MiniLM+LR improves to 0.565 but still trails the keyword baseline (0.879). Real-data fine-tuning is required for trained models to outperform the keyword baseline.

3. **[ASSUMPTION] Hinglish prevalence**: "Indian reviews are predominantly Hinglish" is an assumption. The actual proportion of Hinglish vs pure English vs Hindi varies by platform, city, and business type. The 15% Hinglish in synthetic data is an estimate, not a verified figure.

4. **[ASSUMPTION] Market gap is underserved**: The claim that Hinglish-aware sentiment is "underserved" is based on general knowledge; no systematic market survey was conducted.

5. **No real API integration**: Reviews are uploaded as CSV/JSON. No Google, Meta, or Zomato API integration.

6. **CPU-only**: No GPU acceleration. Embedding model (MiniLM) is optional.

7. **No authentication**: Single-user development mode.

8. **Sarcasm detection**: 80% sarcasm accuracy on gold set. Remaining 20% misclassified because positive keywords appear in sarcastic reviews.

9. **Crisis precision drops on realistic text**: 0.846 precision (vs 1.000 synthetic). Borderline cases trigger false positives.

10. **Name masking coverage**: Limited to ~30 common Indian names.

Fact-Checking Note

All Cynbit company claims in this report were re-verified against https://www.cynbit.in, https://www.cynbit.in/services, https://www.cynbit.in/products, and https://www.cynbit.in/case-studies on October 23, 2026. Revio is confirmed as a real Cynbit product ("AI Reputation & Sentiment Hub"). Crisis detection is our proposed addition, not a confirmed existing Revio feature [ASSUMPTION]. See `docs/research/company_notes.md` for the full fact-check with [VERIFIED]/[ASSUMPTION] markers.

8. Reproducibility

• **Seed**: RANDOM_SEED=42 in `src/config.py` ensures identical output across runs

• **Dependencies**: Core in `requirements.txt`; optional (MiniLM, OpenAI) in `requirements-optional.txt`

• **Commands**: `make data` generates data; `make train` trains models; `make eval` runs evaluation

• **Tests**: `pytest tests/ -v` runs 44 unit tests (all passing)

• **Fresh clone**: `pip install -r requirements.txt && make data && make eval` works in < 5 minutes

9. Roadmap: 30/60/90 Days

Phase 1 (Days 1-30): Real Data Pilot

• Acquire 10,000 real reviews from 5 Revio customer locations

• Label 500 reviews with ground truth (sentiment, aspects, crisis)

• Train TF-IDF + LR baseline and MiniLM + classifier heads on real data

• **Expected outcome**: Trained models should outperform keyword baseline once trained on real data (as opposed to synthetic), closing the 0.448/0.565 gold gap

• Report: Real vs. synthetic accuracy gap analysis

Phase 2 (Days 31-60): Shadow Mode

• Deploy in shadow mode to 2-3 customer locations

• System runs in parallel, captures predictions, human experts validate

• Drift monitoring: Compare metric distributions over time

• A/B testing: Auto-reply vs. human reply comparison

• Fine-tune models on accumulated real data

Phase 3 (Days 61-90): Production Rollout

• Deploy to 5 production locations with human approval gate

• Monitor KPIs: approval rate, customer satisfaction, crisis response time

• Establish MLOps pipeline: weekly retraining on new data

• Customer feedback loop: Survey auto-approved replies

• Operational runbook: On-call procedures, escalation protocols

10. Demo Video Script (2-3 minutes)

1. **Intro (15s)**: "Revio v2 upgrades Cynbit's reputation engine with Hinglish-aware sentiment and grounded replies"

2. **Data generation (15s)**: Show `python data/generate_reviews.py` generating 3,000 reviews

3. **Dashboard (45s)**: Open Streamlit dashboard, load sample data, show sentiment distribution, crisis queue, aspect analysis

4. **API demo (30s)**: Send a Hinglish review to `/analyze` endpoint, show JSON response with aspects, crisis, reply

5. **Crisis handling (30s)**: Send a crisis review ("found glass in food"), show escalation and human review routing

6. **Evaluation (30s)**: Show evaluation metrics, all targets passed

7. **Closing (15s)**: "Synthetic data PoC, 30/60/90-day plan for real-data pilot. Thank you."

Technical approach for Cynbit Technologies AI/ML Intern Assessment. Synthetic evaluation scores are optimistic; manually-authored gold set scores represent honest generalisation. Gold reviews were manually authored with AI assistance, NOT real customer reviews.